Jerry Chen

Waterloo, Ontario
226-337-9007 Email Linkedin Github

Technical Skills

Languages: Python, Java, C++
Software and Tools: Github, Solidworks, Onshape, Raspberry Pi, Linux, Docker, Ubuntu, PyTorch
Concepts: ROS2, Machine Learning, SLAM, Simulation, RViz, Reinforcement Learning

Projects

Scout | *ROS 2 (Jazzy), SLAM, Nav2, Sensor Fusion, Raspberry Pi, Linux*

- I built a ROS 2 rover that maps rooms by itself via **ROS2** using an Elegoo car kit, an **RaspberryPi** and a **LIDAR**
- Wrote **lidar** only **odometry** via ICP scan matching for a car with no wheel encoders or IMU, accurate to about 2 cm.
- Used IR and ultrasonic sensors to detect obstacles the LIDAR can't see, for greater obstacle avoidance.

Rover Fleet | *Gazebo, Pytorch, Reinforcement Learning*

- Built multi-robot exploration system with realistic limits: limited radio, and individualized robot **SLAM**.
- Used **reinforcement learning** to pick robot goals, mapping buildings 25% faster than industry standard strategy.
- Trained in a custom simulator, then proved the concept in a 3-robot Gazebo simulation, mapping 93–94% of the house.

TronShot | *C++, SolidWorks*

- Engineered a two-axis VEX turret with **C++** firmware to accurately hit targets 5m away.
- Designed chain-driven pitch control, with DFM-optimized 3D printed components. **Rapid prototyping** through multiple mechanical prototypes to eliminate shear failures and gear skipping.
- Developed **C++** control software for actuation, firing automation, sensor-based safety logic, and multi-mode operation.

Experience

Doubler Aerospace **August 2026 - PRESENT**
ML Engineer *Guelph, ON*

- **Developed** a machine learning project to predict equipment failure from historical sensor data, creating a working model to estimate remaining operating life.
- Built a data-driven model for predicting airfoil performance, enabling rapid evaluation of different designs and identification of improved configurations.
- Developed a **computer vision** project to detect aircraft in satellite imagery, creating an automated system for identifying and locating objects in images.

WATonomous **September 2025 - PRESENT**
Design Team Member *Waterloo, ON*

- Designed and implemented a custom 3D Nav2 navigation plugin in **ROS2**, supporting **autonomous** pathfinding capabilities.
- Working with **docker** in a containerized environment with a **CI/CD** workflow.
- Implemented vehicle localization utilizing **Extended Kalman Filters** for real-fused state estimation. (EVE)

All Equip LTD **May 2026 - August 2026**
Engineering Assistant *St. Mary's, ON*

- Modeled complete B-train trailer assemblies in **CAD**, developing accurate layouts and component interfaces to support engineering and manufacturing.
- Produced **800+** engineering drawings for shop floor technicians, ensuring manufacturing and assembly specifications were clear and standardized
- Managed and maintained the company's **ERP** system, organizing engineering and production data to improve accessibility and workflow efficiency.

Education

University of Waterloo **September 2025 – PRESENT**
BASc Mechatronics Engineering *Waterloo, On*